HireSense AI

System Architecture, Technical Workflows & Stack Interaction Guide

Enterprise Platform Specification
Version: 2.4.0 (Production Release)
Status: Verified & Operational

1. Executive Architectural Overview

HireSense AI is a full-stack, enterprise-grade Talent Acquisition Assistant built to automate and streamline the end-to-end recruitment lifecycle. The system is engineered using a decoupled client-server architecture designed for zero latency, data privacy compliance, high parsing accuracy, and transparent algorithmic explainability.

> **Core Mission & Guiding Principles**
> Unlike traditional keyword-matching applicant tracking systems (ATS), HireSense AI combines deterministic text extraction, transparent 100-point rubric scoring, verified evidence citation, human-in-the-loop recalibration audit trails, and ethical AI safeguards that completely blind personal demographic attributes from candidate scoring.

High-Level Architectural Blueprint

System Layer	Technology Core	Primary Responsibility & Operations
Presentation Layer	React 18, TypeScript, Tailwind CSS, Lucide Icons	Responsive enterprise HR dashboard, guided 7-stage workflow stepper, interactive scoring sliders, modals, and real-time state feedback.
Build & Dev Tooling	Vite 8, TypeScript Compiler (tsc), ES Modules	Sub-second hot module replacement (HMR), tree-shaking, production bundle optimization, and reverse proxy forwarding (`/api` → `:5000`).
Application Server	Node.js (v22.x), Express 4.x, Multer	RESTful API routing, multipart/form-data stream handling, JSON payload validation, CORS management, and error orchestration.
Document Parsing	PDF.js / pdf-parse, Mammoth (.docx), Plaintext	In-memory binary buffer parsing, text normalization, and structured extraction of job descriptions and candidate resumes.
AI & Inference Engine	Google Gemini 1.5/2.0 API & Deterministic Fallback Engine	Structured JSON schema extraction, 100-point multi-factor candidate evaluation, evidence quote extraction, and contextual interview question generation.
Persistence & State	HTML5 Browser Session Storage & Memory Caches	Client-side persistence of active requisitions, evaluated candidates, live screening interview scores, and immutable recalibration audit logs.

2. Technology Stack Deep Dive: How Each Component Works

React 18 & TypeScript

The UI uses functional components with custom React hooks and Context API (`RecruitmentContext`). TypeScript

Tailwind CSS (Enterprise Palette)

Tailwind CSS provides a refined design system featuring a deep slate navy sidebar (`#0f172a`), subtle purple accents

provides strict compile-time type safety across candidate models, scoring rubrics, and API response contracts.

(`#7c3aed`), clean typography, glassmorphism cards, and responsive data tables.

Node.js & Express REST Backend

Express runs as an asynchronous microservice on port 5000. It handles file uploads in memory via Multer, dispatches extraction jobs to parsing services, and coordinates scoring calculations.

Google Gemini GenAI SDK

Integrated via `@google/generative-ai` with `responseMimeType: "application/json"`. Prompts enforce strict output schemas with zero hallucination guardrails.

PDF.js / pdf-parse

Decodes incoming PDF binary streams into UTF-8 text buffers. Extracts raw document layouts without external system dependencies or cloud upload risks.

Mammoth DOCX Parser

Reads Office Open XML (DOCX) files directly from in-memory byte arrays. Converts word processor paragraphs and bullet hierarchies into sanitized plaintext.

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Complete 7-Stage Talent Acquisition Workflow Lifecycle

Workflow Execution Manual
Stage-by-Stage Operational Walkthrough

3. The Seven Mandatory Recruitment Workflows

1 Workflow 1: Recruiter Workspace Authentication & Session Management

Objective: Establish a verified recruiter session and load active requisitions, recruiter metrics, and role permissions.

- **Mechanism:** The recruiter selects their enterprise profile (e.g., *Sarah Jenkins – Lead Technical Recruiter* or *Karthik Subramanian – Senior Talent Acquisition Specialist*) or submits corporate credentials.
- **Data Handling:** The Express endpoint (`POST /api/auth/login`) validates the session, issues a session token, and returns the recruiter's active requisitions and placement targets.
- **Persistence:** The user session is automatically mirrored to browser `sessionStorage`, ensuring state survives navigation, tab switches, and page refreshes.

2 Workflow 2: Job Description Upload, Extraction & Interactive Editing

Objective: Ingest job descriptions across heterogeneous formats (PDF, DOCX, pasted text) and extract a normalized, editable schema.

- **Ingestion:** Accepts drag-and-drop `.pdf`, `.docx`, or raw text input. Also provides a 1-click benchmark role import (*Senior Full Stack & AI Solutions Engineer*).
- **Extraction Engine:** Backend parses the binary stream via `pdf-parse` or `mammoth`. Gemini AI (or the local deterministic NLP extractor) structures the text into: Role Title, Department, Location, Experience Range, Mandatory Technical Skills, Preferred Competencies, and Responsibilities.
- **Recruiter Editability:** Recruiters can enter edit mode to add, remove, or modify required skills and duties prior to ranking, guaranteeing the rubric perfectly matches hiring manager criteria.

3 Workflow 3: Multi-Resume Ingestion & Binary Document Extraction

Objective: Bulk-process candidate resumes concurrently with status tracking, metadata extraction, and candidate pool management.

- **Upload Pipeline:** Multer receives multipart file arrays in memory (up to 20 files per batch). The system extracts full text, calculates file sizes, and parses candidate names and career timelines.
- **Benchmark Candidate Pool:** Features 5 pre-screened synthetic profiles representing realistic talent diversity: *Aravind Sharma* (Top match, ~92), *Priya Venkat* (Backend/Cloud, ~84), *Marcus Chen* (ML/Data, ~72), *Neha Kapoor* (Frontend/UI, ~58), and *David Miller* (Legacy Java, ~42).
- **Status Tracking:** Real-time processing indicators transition from `Uploading` to `Extracting` to `Parsed & Ready`.

4 Workflow 4: Transparent 100-Point AI Candidate Ranking & Recalibration Audit

Objective: Perform unbiased multi-factor evaluation with verbatim evidence citations and human-in-the-loop override capabilities.

- **100-Point Scoring Rubric:**
  • *Technical Skills Match (40 pts):* Direct evaluation of mandatory (30 pts) and preferred (10 pts) stack proficiency.
  • *Experience Depth (25 pts):* Total verified engineering years relative to role seniority requirements.
  • *Project Relevance & Scale (20 pts):* Quantifiable impact in high-scale microservices and production AI pipelines.
  • *Education & Foundation (15 pts):* Accredited Engineering/CS degree or equivalent domain foundations.
- **Ethical AI Guarantee:** Candidate name, gender, age, ethnicity, nationality, photo, and marital status have strictly **0% algorithmic weight**.
- **Evidence-Based Explanations:** Exact verbatim quotes from candidate resumes are extracted to substantiate each score dimension.
- **Human-in-the-Loop Recalibration Audit:** Recruiters can challenge an automated score (e.g., if custom Triton inference was under-weighted by initial keyword match), adjust point deltas, submit review notes, and permanently record an immutable audit entry showing Original Score, Corrected Score, and Delta.

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Screening Intelligence, Market Analytics & Talent Sourcing Workflows

Workflow Execution Manual (Part II)
Market Analytics & Sourcing Query Engine

5 Workflow 5: Contextual Screening Interview Questions & Evaluation Rubric

Objective: Equip technical recruiters and hiring managers with structured, high-signal interview questions tailored to the active JD.

- **Structured Composition:** Generates exactly 10 questions:
  • *Technical Architecture & Code:* 4 questions (RAG pipelines, state streaming, database scaling, container security).
  • *Practical Experience & Incidents:* 3 questions (production outage diagnosis, technical debt vs. velocity, inference cost management).
  • *Role & Cultural Fit:* 3 questions (cross-functional communication, junior mentorship, continuous technology vetting).
- **Expected Good-Answer Benchmarks:** Concrete bulleted answer criteria detailing what exceptional candidate responses must include.
- **Interactive Recruiter Scorekeeper:** Includes live 1-to-5 star rating inputs and a phone screening notes drawer for real-time interview documentation.

6 Workflow 6: Public Hiring Market Intelligence & Compensation Percentiles

Objective: Provide actionable salary benchmarks, talent demand metrics, and comparable public job postings.

- **Comparable Public Postings:** Displays active public technology job postings from top tier firms (Razorpay, CRED, Swiggy, Thoughtworks) with role, experience, compensation (INR LPA and USD), location, and direct careers portal URLs.
- **Transparent Verification Badges:** Every benchmark is explicitly tagged as either `[Verified Public Posting]` or `[Estimated Market Benchmark]`.
- **Compensation Distribution:** Visualizes the 25th percentile (₹28 LPA), Median (₹36 LPA), 75th percentile (₹44 LPA), and 90th percentile (₹50+ LPA) salary spread.
- **Compliance Standard:** Operates strictly under open public data standards. Zero unauthorized web scraping or authenticated sessions of LinkedIn or Naukri are performed.

7 Workflow 7: Dual-Engine Sourcing Search String Generator (Naukri & LinkedIn)

Objective: Produce production-ready talent sourcing queries formatted with platform-specific Boolean logic and plain language keyword densities.

- **Naukri Boolean & Plain:** Formatted with `AND`, `OR`, `NOT`, and parentheses grouping optimized for Naukri Resdex keyword matching. Includes comma-separated plain keywords for quick recruiter filters.
- **LinkedIn Boolean & Plain:** Structured with title parameters (`title:"Full Stack Engineer" ...`) and semantic phrases optimized for LinkedIn Recruiter search algorithms.
- **1-Click Clipboard Copy:** Each search query features an instant copy button with visual toast feedback for rapid recruiter sourcing workflows.

4. Data Flow & Inter-Process Architecture

End-to-End Execution Sequence

1. **Ingest:** Client posts binary file payload to Express `/api/jd/parse` or `/api/resumes/upload`.
2. **Extract:** Express streams binary buffer to `parserService.js`, invoking `pdf-parse` or `mammoth.extractRawText()`.
3. **Analyze:** Normalized text is processed by `geminiService.js` or the deterministic rule engine to extract structured JSON entities.
4. **Evaluate:** `rankingService.js` computes category point breakdowns against the 100-point rubric and checks against ethical bias rules.
5. **Audit:** If recruiter adjusts points, `/api/ranking/recalibrate` records delta, reason, timestamp, and audit ID into the persistent log.
6. **Synchronize:** Client receives verified JSON, updates local state, and writes the state snapshot to browser `sessionStorage`.

5. Security, Ethics & Production Readiness

- **Secret Protection:** Gemini API keys are maintained exclusively on the backend in environment variables (`.env`) and are never exposed to the client bundle.
- **Resilient Offline Fallback:** If an API key is missing or encounters network quota limits, HireSense AI automatically fails over to its deterministic evaluation and scoring engines, ensuring 100% operational uptime.
- **Compliance & Privacy:** Zero candidate data or job descriptions are transmitted to unauthorized third parties or persistent external databases.
